Postgraduate Module
MAP55672: Case Studies in HPC
Case 2: GMRES

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Rules

Due Monday 23 March at 10:00pm. Please create a git repository with your solutions to this assignment, and submit this link as a part of a PDF with any responses and plots on Blackboard. Your submission should contain a Matlab Live file or Jupyter notebook file containing all your demonstrations and solutions, and you should create separate files containing function definitions.

Question

In this assignment, you are being asked to implement GMRES and run some tests with it. You should use the textbooks from Saad and from Trefethen and Bau as references. Please submit all three versions of your GMRES code.

1. Write a Matlab code implementing the GMRES algorithm.

   - Begin with a basic impelemention that just solves the least minimisation of

   $$\left\| \underline{H_m} \boldsymbol{y} - \|\boldsymbol{r}_0\| \, \boldsymbol{e}_1 \right\|$$

   - Modify your implementation to use the progressive Givens rotation implementation for the QR-factorisation of $\underline{H_m}$ as described in the lecture and in Saad's textbook.
   - Further modify your implementation so that the residual norm is monitored without actually building the GMRES approximate solution. Your code should run a loop that checks the residual norm approximation and *only* builds the GMRES approximation once the residual norm goes below tolerance.
   - Please test your functions on the small and large test problems from the SuiteSparse database of test matrices.